

Edward Lei

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EDUCATION

Carnegie Mellon University

B.S. in Electrical and Computer Engineering

GPA: 3.91/4.0

Relevant Coursework: Introduction to ASIC and FPGA Chip Design, Structure and Design of Digital Systems, Introduction to Microelectronic Circuits, Introduction to Embedded Systems, Introduction to Computer Systems

Pittsburgh, PA

Aug 2023 – May 2027

PROJECT EXPERIENCE

Custom CMOS Datapath Design & Characterization

Pittsburgh, PA

Designer

March 2026

- Architected and characterized custom negative-edge triggered master-slave flip-flops and combinatorial datapath logic in Cadence Virtuoso, performing transient simulations using Spectre to validate functional logic.
- Evaluated critical timing metrics under a 2.4 GHz clock domain—calculating a 32 ps setup time and 45.16 ps propagation delay to determine a robust setup slack of 339.5 ps (81% of the period)
- Designed full-custom transistor-level layouts for custom master-slave flip-flop datapaths spanning a 54.57 μm^2 area in Cadence Virtuoso.
- Executed comprehensive physical verification, successfully achieving 0 DRC and LVS violations, followed by parasitic extraction to re-evaluate timing metrics and verify post-layout robustness.

SystemVerilog VGA Slot Machine Chip Design & Tapeout

Pittsburgh, PA

Designer

January 2026 – May 2026

- Architected a hardware-accelerated VGA slot machine controller using SystemVerilog, implementing finite state machines (FSMs) to dynamically regulate display cycle speeds and game rendering states.
- Executed an open-source RTL-to-GDSII physical design flow utilizing Yosys for logic synthesis, nextpnr for placement and routing, OpenLane/OpenROAD for automated hardening, and fujprog for hardware validation.
- Successfully taped out a custom silicon design onto a multi-project wafer shuttle via Tiny Tapeout, managing design constraint optimization and physical verification.

FPGA Physical Design & Performance Tuning

Pittsburgh, PA

Designer

February 2026

- Optimized the critical path of a 32-bit arithmetic unit by implementing multi-stage pipelining, utilizing Static Timing Analysis (STA) to calculate the maximum clock frequency and balance logic vs. routing delays.
- Engineered high-throughput architectures on Lattice ECP5 and iCE40 FPGAs by explicitly instantiating MULT18X18D DSP blocks and performing manual floor-planning via Python-scripted constraints in NEXTPNR to minimize Look-Up Table (LUT) overhead.

PROFESSIONAL EXPERIENCE

JACS Solutions

Linthicum Heights, MD

Hardware & Automation Engineering Intern

May 2026 - July 2026

- Automated hardware verification and testing workflows by developing custom Python testing scripts (leveraging pyserial, subprocess, and regular expressions), parsing complex Excel test cases and serial log files to streamline validation for 4G/5G Ethernet dongles and Android MDM software.
- Performed hands-on hardware debugging, identifying bricked peripheral units and reviving them by shorting internal PCB test points and flashing firmware utilizing QFIL software utilities.
- Engineered an automated IP passthrough testing script that integrates terminal result parsing with public IPv4/IPv6 APIs to validate carrier network compatibility across Verizon, AT&T, and T-Mobile SIM cards.

Microelectronic Circuits (18-320)

Pittsburgh, PA

Teaching Assistant

Aug 2026 – Present

- Support instruction covering MOSFET device physics and fabrication, analog/digital CMOS logic design, clocked storage elements, single-stage and differential amplifiers, frequency response analysis, and op-amp based analog filter design.
- Guide student teams through a semester-long CAD-based project designing and laying out a capacitive sensing system analogous to touchscreen sensor architectures, using Cadence for schematic capture and layout.
- Assist students with hands-on lab work spanning discrete/SSI/MSI circuit implementations and industry-standard CAD design flows, using SLICE/SPICE for circuit simulation and verification.

ADDITIONAL INFORMATION

Skills: C, C++, Python, System Verilog, RTL Design, OpenLane (ASIC Flow), Yosys, Synopsys VCS, Verilator, CocoTB, Cadence Virtuoso (Spectre/SPICE), NextPNR